Yuni Wu

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Summary

AI Systems Engineer focused on traceable LLM workflows, agent runtime reliability, and evaluation-first product behavior. Builds Python/TypeScript systems where tool routing, evidence aggregation, streaming execution, feedback loops, and cloud orchestration make AI outputs easier to test, debug, and trust.

Experience

Graduate Research Assistant Sept 2025 – May 2026
CAIRO Lab, University of Colorado Boulder — Robotics and embodied AI research lab

- Improved autonomous waypoint prediction to 0.11 MAE and $R^2 = 0.62$ by fine-tuning PyTorch/torchvision ResNet-18 behavior-cloning policies on 97K+ robot state-action samples with weighted regression losses, transfer learning, and cross-validation.
- Converted ROS/rosbag telemetry into reproducible datasets and validated behavior on physical Unitree Go1 robots through latency checks, stability validation, inference-time ensembling, smoothing, and safety monitoring.

Machine Learning Engineer Intern May 2025 – Aug 2025
MyEdMaster LLC — AI-driven healthcare and education technology company

- Verified 0.92 F1-score on a 22-class medical-image classification workflow by evaluating ResNet-based models across 50K+ imbalanced samples with repeatable validation scripts.
- Reduced per-image inference latency by 5× (0.15s to 0.03s) by comparing ResNet architectures, measuring deployment tradeoffs, and selecting faster model variants without dropping reliability checks.

Software Engineer June 2021 – Aug 2022
Shopee Co. Ltd. — Major Southeast Asian e-commerce platform

- Reduced manual QA workload by 80% with Go API validation, SQL checks, and anomaly-detection workflows across 800+ production marketplace scenarios.
- Improved operational reporting speed by 78% through regression-monitoring pipelines and reduced query latency by 15% through distributed ETL and GCP data-access optimizations.

Projects

Stock Intelligence Copilot — Agentic Stock Research Platform [demo]

- Enabled traceable multi-step research across market data, fundamentals, news, earnings, and portfolio signals through a FastAPI/React/Next.js agentic analysis platform.
- Orchestrated LangChain/LCEL-supported execution for intent classification, planning, tool routing, evidence aggregation, and response synthesis across research, explain, trade, watchlist, and portfolio workflows.
- Improved runtime traceability and debugging by implementing FastAPI SSE streaming, structured execution schemas, route adapters, per-tool error capture, fallback-safe API presentation, and automated tests across runtime execution, agent schemas, tool contracts, and streaming response behavior.

EmotionFlow — LLM Preference and Recommendation Evaluation System [demo]

- Operationalized LLM response evaluation by collecting A/B preference choices, helpfulness ratings, optional rationales, and recommendation feedback through MongoDB-backed logging and Dash analytics.
- Enabled offline preference-learning experiments by exporting pairwise response comparisons into DPO-style JSONL and computing preference/evaluation aggregate metrics without claiming model training.

NailAI — Cloud-Native ML Inference Platform [demo]

- Recorded 0 failures in a load test at 200 concurrent users with p95 latency of 6.8s through Cloud Run services, Pub/Sub job routing, retry handling, and distributed backend/worker coordination.
- Improved inference observability by instrumenting request, job, and prediction metadata for monitoring, downstream analysis, and reliability investigation across services.

Technical Skills

Languages: Python, TypeScript, SQL, Go
AI Systems: OpenAI API integration, LangChain/LCEL workflows, LangGraph, RAG, intent classification, tool routing, evidence aggregation, structured response synthesis, feedback-driven evaluation, preference capture, DPO-style export
Backend & Cloud: FastAPI, Flask, React, Next.js, REST APIs, Server-Sent Events (SSE), GCP Cloud Run, Pub/Sub, Cloud Storage, Docker, BigQuery, Kubernetes, asynchronous job orchestration
Reliability & Testing: structured execution schemas, observability-oriented metadata, fallback-safe APIs, regression validation, automated regression testing, QA automation, Git, pytest/unittest, Playwright end-to-end tests, load-test scaffolding

Education

University of Colorado Boulder — MS, Data Science | GPA: 3.76/4.0 May 2026
Soochow University — BS, Data Science | GPA: 3.7/4.0 June 2021